

Section 4.3

$$\begin{aligned}
 21 \quad E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy = \int_{-\infty}^0 y \cdot 0 dy + \int_0^1 y \cdot \left[\left(\frac{3}{2}\right) y^2 + y \right] dy + \int_1^{\infty} y \cdot 0 dy \\
 &= \int_0^1 \left[\frac{3}{2} y^3 + y^2 \right] dy = \left[\frac{3}{8} y^4 + \frac{1}{3} y^3 \right]_0^1 = \frac{17}{24} \text{ hrs}
 \end{aligned}$$

$$\begin{aligned}
 V(Y) &= E(Y^2) - [E(Y)]^2 \\
 E(Y^2) &= \int_0^1 y^2 \cdot \left[\left(\frac{3}{2}\right) y^2 + y \right] dy = \int_0^1 \left[\frac{3}{2} y^4 + y^3 \right] dy = \left[\frac{3}{10} y^5 + \frac{1}{4} y^4 \right]_0^1 = \frac{11}{20}
 \end{aligned}$$

$$V(Y) = \frac{11}{20} - \left(\frac{17}{24}\right)^2 = \frac{139}{2880} = .0483 \text{ hrs}^2$$

$$27 \quad E(W) = E(5 - .5Y) = E(5) - E(.5Y) = 5 - .5 E(Y)$$

Recall $E(Y)$ from #21 is $\frac{17}{24}$

$$E(W) = 5 - .5 \left(\frac{17}{24}\right) = \$4.65$$

$$V(W) = (-.5)^2 V(Y) = 0.25 \cdot \frac{139}{2880} = .0121 \text{ dollar}^2$$

$$\begin{aligned}
 29 \quad E(Y) &= \int_{-\infty}^{\infty} y \cdot f(y) dy = \int_{-\infty}^{59} y \cdot 0 dy + \int_{59}^{61} y \cdot \frac{1}{2} dy + \int_{61}^{\infty} y \cdot 0 dy \\
 &= \int_{59}^{61} \frac{1}{2} y dy = \left[\frac{1}{4} y^2 \right]_{59}^{61} = \frac{61^2}{4} - \frac{59^2}{4} = 60 \text{ degrees}
 \end{aligned}$$

$$\begin{aligned}
 V(Y) &= E(Y^2) - [E(Y)]^2 \\
 E(Y^2) &= \int_{59}^{61} y^2 \cdot \frac{1}{2} dy = \left[\frac{1}{6} y^3 \right]_{59}^{61} = \frac{61^3}{6} - \frac{59^3}{6} = \frac{10801}{3}
 \end{aligned}$$

$$V(Y) = \frac{10801}{3} - 60^2 = \frac{1}{3} \text{ degrees}^2$$